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Research paper

Hyper Dual Quaternions representation of rigid bodies kinematics

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Hyper-Dual Quaternions, HDQ, are first introduced in this paper. Utilizing the Hyper-Dual Numbers, HDN, Dual Quaternions, DQ, and hyper-dual angle, the general expression of HDQ is obtained. Rigid single- and multi-body kinematics such as in a serial robot kinematics, are then expressed in HDQ form. Taking advantage of the dual numbers' "automatic differentiation" feature, HDQ encompasses both body pose (translational and rotational) and body velocities in a compact form with no need for further pose differentiation.

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1. Introduction

In 1843 Hamilton invented the quaternions that perform rotation of vectors in a three-dimensional space. Then, in 1870, Clifford introduced the dual numbers which he applied to quaternion and named bi-quaternions that combine both rotation and translation in one expression [1].

Today, bi-quaternions, commonly named Dual Quaternion (DQ), are widely used in rigid body kinematics and dynamics [2–6], computer graphics [7–12], multirotor attitude and trajectory tracking [13–16], general multirotor control purposes [17–21], and for the translation and attitude synchronization of multiple rigid bodies [22,23].

Recently Hyper-Dual Numbers (HDN) were introduced by Fike and Alonso [24–27]. As evident from those investigations, the purpose of Fike and Alonso in introducing the HDN was to perform second-order numerical differentiation that leads to smaller numerical (subtractive and cancellation) errors as well as to reduced computational time. Then, in 2015 and 2017, HDN were utilized by the authors of the present paper, to represent rigid body's kinematics and dynamics [28,29]. Taking advantage of the "automatic differentiation" feature of the HDN, rigid body equations-of-motion were derived in a compact form with no need for further differentiation.

The aim of the present investigation is to introduce the Hyper-Dual Quaternions (HDQ), composed of hyper-dual hyper-complex numbers. This is achieved by replacing every real number in a quaternion by the associated HDN. We then proceed to represent a rigid body kinematics in HDQ form.

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List of symbols

a	scalar
$\hat{a}$	dual number
$\overset{\frown}{a}$	hyper-Dual number
$f(\overset{\frown}{a})$	function of a Hyper-Dual number
$\overset{\frown}{\boldsymbol{v}}$	Hyper-Dual vector
ε	dual unit
ε^*	Hyper-Dual unit
$\boldsymbol{q}$	quaternion
$\hat{\boldsymbol{q}}$	dual Quaternion
$\overset{\frown}{\boldsymbol{q}}$	Hyper-Dual Quaternion
$\hat{\boldsymbol{\xi}}$	dual twist
$\boldsymbol{\omega}$	angular velocity
$\hat{\theta}$	dual angle
$\overset{\frown}{\theta}$	Hyper-Dual angle

2. Mathematical formulation of Hyper-Dual Quaternions

2.1. Arithmetic operations of Dual Quaternions (DQ)

We next give the elementary arithmetic operations of DQ, the reader is referred to Appendix A for a list of elementary arithmetic operations necessary for quaternions.

For a DQ, $\hat{\boldsymbol{q}} = \boldsymbol{q}_r + \varepsilon\boldsymbol{q}_d$, where $\boldsymbol{q}_r$ is named the real part, $\boldsymbol{q}_d$ is named the dual part and ε is a dual unit with the property $\varepsilon^2 = 0$, the elementary arithmetic operations are [8]:

Scalar Multiplication (by a scalar s)

$$s\hat{\boldsymbol{q}} = s\boldsymbol{q}_r + \varepsilon s\boldsymbol{q}_d \tag{1}$$

Addition

$$\hat{\boldsymbol{q}}_1 + \hat{\boldsymbol{q}}_2 = \boldsymbol{q}_{r1} + \boldsymbol{q}_{r2} + \varepsilon(\boldsymbol{q}_{d1} + \boldsymbol{q}_{d2}) \tag{2}$$

Multiplication

$$\hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{q}}_2 = \boldsymbol{q}_{r1} \otimes \boldsymbol{q}_{r2} + \varepsilon(\boldsymbol{q}_{r1} \otimes \boldsymbol{q}_{d2} + \boldsymbol{q}_{d1} \otimes \boldsymbol{q}_{r2}) \tag{3}$$

where quaternion multiplication follows Hamilton's multiplication rule of the vector components $i^2 = j^2 = k^2 = ijk = -1$. Other quaternion multiplication formulae exist such as representing the quaternion elements in a matrix form [30] which also follows Hamilton's quaternions multiplication rule by matrix multiplication and leads to the same result.

Conjugate

$$\hat{\boldsymbol{q}}^* = \boldsymbol{q}_r^* + \varepsilon\boldsymbol{q}_d^* \tag{4}$$

Magnitude – For a DQ, the magnitude is defined as the DQ multiplied by its conjugate,

$$\left\|\hat{\boldsymbol{q}}\right\| = \sqrt{\hat{\boldsymbol{q}}\hat{\boldsymbol{q}}^*} \tag{5}$$

Since a DQ is a dual number with quaternion entries, we use the definition of the square root of a dual number to compute the square root of the DQ

$$\sqrt{\boldsymbol{q}_r + \varepsilon\boldsymbol{q}_d} = \sqrt{\boldsymbol{q}_r} + \varepsilon\frac{\boldsymbol{q}_d}{2\sqrt{\boldsymbol{q}_r}}, \quad \boldsymbol{q}_r \neq 0.$$

Hence, for a given DQ, $\hat{\boldsymbol{q}} = \boldsymbol{q}_r + \varepsilon\boldsymbol{q}_d$, and its conjugate, $\hat{\boldsymbol{q}}^* = \boldsymbol{q}_r^* + \varepsilon\boldsymbol{q}_d^*$,

$$\hat{\boldsymbol{q}}\hat{\boldsymbol{q}}^* = (\boldsymbol{q}_r + \varepsilon\boldsymbol{q}_d)\left(\boldsymbol{q}_r^* + \varepsilon\boldsymbol{q}_d^*\right) = \boldsymbol{q}_r\boldsymbol{q}_r^* + \varepsilon\left(\boldsymbol{q}_r\boldsymbol{q}_d^* + \boldsymbol{q}_d\boldsymbol{q}_r^*\right) = \|\boldsymbol{q}_r\|^2 + \varepsilon\left(\boldsymbol{q}_r\boldsymbol{q}_d^* + \boldsymbol{q}_d\boldsymbol{q}_r^*\right)$$

and the Magnitude of the DQ is

$$\left\|\hat{\boldsymbol{q}}\right\| = \sqrt{\hat{\boldsymbol{q}}\hat{\boldsymbol{q}}^*} = \sqrt{\|\boldsymbol{q}_r\|^2 + \varepsilon\left(\boldsymbol{q}_r\boldsymbol{q}_d^* + \boldsymbol{q}_d\boldsymbol{q}_r^*\right)} = \sqrt{\|\boldsymbol{q}_r\|^2} + \varepsilon\frac{\boldsymbol{q}_r\boldsymbol{q}_d^* + \boldsymbol{q}_d\boldsymbol{q}_r^*}{2\sqrt{\|\boldsymbol{q}_r\|^2}} = \|\boldsymbol{q}_r\| + \varepsilon\frac{\boldsymbol{q}_r\boldsymbol{q}_d^* + \boldsymbol{q}_d\boldsymbol{q}_r^*}{2\|\boldsymbol{q}_r\|}$$

Where for a unit DQ, that has a magnitude of 1, we use the unit condition

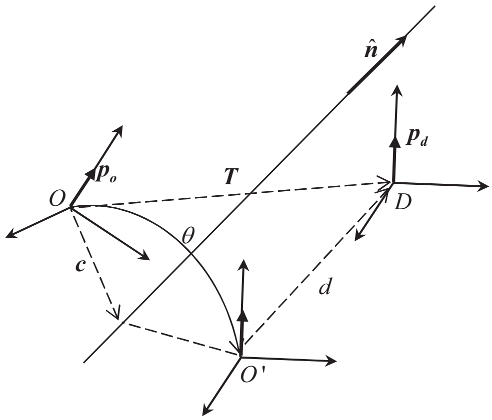

Fig. 1. Screw motion, along a unit vector axis $\boldsymbol{n}$.

$$\begin{aligned} \|\boldsymbol{q}_r\| &= 1 \\ \boldsymbol{q}_r\boldsymbol{q}_d^* + \boldsymbol{q}_d\boldsymbol{q}_r^* &= 0 \end{aligned} \tag{6}$$

Inverse

$$\hat{\boldsymbol{q}}^{-1} = \frac{1}{\|\hat{\boldsymbol{q}}\|^2} \otimes \hat{\boldsymbol{q}}^* \tag{7}$$

Where $q_r \neq 0$, and for a unit DQ, the inverse is $\hat{\boldsymbol{q}}^{-1} = \hat{\boldsymbol{q}}^*$.

2.2. Dual angle and the unit Dual-Quaternion

Next the dual quaternion representation of rigid body's kinematics is defined. Let a coordinate frame D undergoes translation **T** and rotation θ with respect to frame O, (see Fig. 1).

Using Study's dual angle [31], the coordinates of frame D is given by the DQ [32]:

$$\begin{aligned} \hat{\boldsymbol{q}} = \boldsymbol{q} + \varepsilon\frac{1}{2}\boldsymbol{q}\otimes\boldsymbol{T} &= \left[C\left(\frac{\theta}{2}\right), S\left(\frac{\theta}{2}\right)\boldsymbol{n}\right] + \frac{\varepsilon}{2}\left[C\left(\frac{\theta}{2}\right), S\left(\frac{\theta}{2}\right)\boldsymbol{n}\right]\otimes[0, \boldsymbol{T}] \\ &= \left[C\left(\frac{\theta}{2}\right), S\left(\frac{\theta}{2}\right)\boldsymbol{n}\right] + \frac{\varepsilon}{2}\left[-S\left(\frac{\theta}{2}\right)\boldsymbol{n}^T \boldsymbol{.}\, \boldsymbol{T},\ C\left(\frac{\theta}{2}\right)\boldsymbol{T} + S\left(\frac{\theta}{2}\right)\boldsymbol{n}\times\boldsymbol{T}\right] \end{aligned} \tag{8}$$

2.3. Mathematical formulation of Hyper-Dual numbers

Hyper-dual numbers (HDN) were first introduced by Fike et al. [24–27]. In [28] Hyper-dual numbers were augmented to better represent rigid body's kinematics.

An HDN, $\widehat{x}$, is a number consisting of four real numbers (x_1, x_2, x_3, x_4) and two dual units, ε, ε^*, with the following multiplication rules:

$$\varepsilon^2 = \varepsilon^{*2} = (\varepsilon\,\varepsilon^*)^2 = 0 \tag{9}$$

$$\varepsilon\,,\ \varepsilon^*,\ \varepsilon\,\varepsilon^* \neq 0 \tag{10}$$

Where $\widehat{x}$ is:

$$\widehat{x} = x_1 + \varepsilon x_2 + \varepsilon^*(x_3 + \varepsilon\, x_4) \tag{11}$$

In this paper dual numbers are marked as $\hat{x}$ and HDN are marked as $\widehat{x}$.

2.4. Arithmetic operations of Hyper-Dual Quaternions

A HDQ, is a quaternion with HDN components,

$$\widehat{\boldsymbol{q}} = \left[\widehat{a}, \widehat{\boldsymbol{v}}\right] = \widehat{a} + \widehat{x}i + \widehat{y}j + \widehat{z}k$$

yielding a hyper-dual hyper-complex number, where $\widehat{a}$ is the scalar part of HDN, and $\widehat{\boldsymbol{v}}$ is it's vector part (hyper-dual vectors).

The dual units ε, ε^* commute with the quaternion units, for example $i\varepsilon = \varepsilon i$.

The elementary arithmetic operations of HDQ, $\widehat{\boldsymbol{q}} = \hat{\boldsymbol{q}}_d + \varepsilon^* \hat{\boldsymbol{q}}_{hd}$, where $\hat{\boldsymbol{q}}_d$ is a DQ composing the dual part of $\widehat{\boldsymbol{q}}$, and $\hat{\boldsymbol{q}}_{hd}$ is a DQ composing the hyper-dual part of $\widehat{\boldsymbol{q}}$, are:

Scalar Multiplication (by scalar s)

$$s\widehat{\boldsymbol{q}} = s\hat{\boldsymbol{q}}_d + s\varepsilon^* \hat{\boldsymbol{q}}_{hd} \tag{12}$$

Addition

$$\widehat{\boldsymbol{q}}_1 + \widehat{\boldsymbol{q}}_2 = \hat{\boldsymbol{q}}_{d1} + \hat{\boldsymbol{q}}_{d2} + \varepsilon^* \left(\hat{\boldsymbol{q}}_{dh1} + \hat{\boldsymbol{q}}_{dh2} \right) \tag{13}$$

Multiplication - For two quaternions, $\widehat{\boldsymbol{q}}_1$, $\widehat{\boldsymbol{q}}_2$, with HDN components,

$$\widehat{\boldsymbol{q}}_1 = \left[\widehat{a}_1 \, , \, \widehat{\boldsymbol{v}}_1 \right] = \widehat{\boldsymbol{q}}_{d1} + \varepsilon^* \widehat{\boldsymbol{q}}_{hd1}$$

$$\widehat{\boldsymbol{q}}_2 = \left[\widehat{a}_2 \, , \, \widehat{\boldsymbol{v}}_2 \right] = \widehat{\boldsymbol{q}}_{d2} + \varepsilon^* \widehat{\boldsymbol{q}}_{hd2}$$

HDQ multiplication is:

$$\widehat{\boldsymbol{q}}_1 \otimes \widehat{\boldsymbol{q}}_2 = \left(\hat{\boldsymbol{q}}_{d1} + \varepsilon^* \hat{\boldsymbol{q}}_{hd1} \right) \otimes \left(\hat{\boldsymbol{q}}_{d2} + \varepsilon^* \hat{\boldsymbol{q}}_{hd2} \right) = \hat{\boldsymbol{q}}_{d1} \otimes \hat{\boldsymbol{q}}_{d2} + \varepsilon^* \left(\hat{\boldsymbol{q}}_{d1} \otimes \hat{\boldsymbol{q}}_{hd2} + \hat{\boldsymbol{q}}_{hd1} \otimes \hat{\boldsymbol{q}}_{d2} \right) \tag{14}$$

Conjugate– The conjugate of a HDQ is the extension of the conjugate of a dual quaternion, that is

$$\widehat{\boldsymbol{q}}^* = \hat{\boldsymbol{q}}_d^* + \varepsilon^* \hat{\boldsymbol{q}}_{hd}^* \tag{15}$$

HDQ Magnitude – A HDQ multiplied by its conjugate gives the magnitude squared, hence:

$$\left\| \widehat{\boldsymbol{q}} \right\| = \sqrt{\widehat{\boldsymbol{q}} \widehat{\boldsymbol{q}}^*} \tag{16}$$

Where the square root of the HDQ, $\widehat{\boldsymbol{q}} = \boldsymbol{q}_r + \varepsilon \boldsymbol{q}_d + \varepsilon^* (\boldsymbol{q}_{hr} + \varepsilon \boldsymbol{q}_{hd})$, is computed from the definition of the square root of an HDN [33]

$$\sqrt{\widehat{\boldsymbol{q}}} = \sqrt{\boldsymbol{q}_r} + \frac{\varepsilon \boldsymbol{q}_d}{2\sqrt{\boldsymbol{q}_r}} + \varepsilon^* \left(\frac{\boldsymbol{q}_{hr}}{2\sqrt{\boldsymbol{q}_r}} + \varepsilon \left(\frac{\boldsymbol{q}_{hd}}{2\sqrt{\boldsymbol{q}_r}} + \frac{\boldsymbol{q}_d \boldsymbol{q}_{hr}}{4\sqrt{\boldsymbol{q}_r}^3} \right) \right) \quad , \boldsymbol{q}_r \neq 0 \, .$$

For a given HDQ, $\widehat{\boldsymbol{q}} = \hat{\boldsymbol{q}}_d + \varepsilon^* \hat{\boldsymbol{q}}_{hd}$, and its conjugate, $\widehat{\boldsymbol{q}}^* = \hat{\boldsymbol{q}}_d^* + \varepsilon^* \hat{\boldsymbol{q}}_{hd}^*$,

$$\widehat{\boldsymbol{q}} \widehat{\boldsymbol{q}}^* = \left(\hat{\boldsymbol{q}}_d + \hat{\boldsymbol{q}}_{hd} \varepsilon^* \right) \left(\hat{\boldsymbol{q}}_d^* + \varepsilon^* \hat{\boldsymbol{q}}_{hd}^* \right) = \hat{\boldsymbol{q}}_d \hat{\boldsymbol{q}}_d^* + \varepsilon^* \left(\hat{\boldsymbol{q}}_d \hat{\boldsymbol{q}}_{hd}^* + \hat{\boldsymbol{q}}_{hd} \hat{\boldsymbol{q}}_d^* \right) = \left\| \hat{\boldsymbol{q}}_d \right\|^2 + \varepsilon^* \left(\hat{\boldsymbol{q}}_d \hat{\boldsymbol{q}}_{hd}^* + \hat{\boldsymbol{q}}_{hd} \hat{\boldsymbol{q}}_d^* \right)$$

Hence, for a unit HDQ, that has a magnitude of 1, we define the HDQ unit condition as

$$\|\boldsymbol{q}_r\| = 1$$
$$\boldsymbol{q}_r \boldsymbol{q}_d^* + \boldsymbol{q}_d \boldsymbol{q}_r^* = 0$$
$$\hat{\boldsymbol{q}}_d \hat{\boldsymbol{q}}_{hd}^* + \hat{\boldsymbol{q}}_{hd} \hat{\boldsymbol{q}}_d^* = 0$$

3. Hyper-Dual Quaternions representation of kinematics

Given a quaternion,

$$\boldsymbol{q} = \left[C\left(\frac{\theta}{2} \right), S\left(\frac{\theta}{2} \right) \boldsymbol{n} \right] \tag{17}$$

a dual quaternion is defined by

$$\boldsymbol{q} = \left[C\left(\frac{\hat{\theta}}{2} \right), S\left(\frac{\hat{\theta}}{2} \right) \hat{\boldsymbol{n}} \right] \tag{18}$$

Where $\hat{\theta}$ is Study's dual angel [31] and $\hat{\boldsymbol{n}}$ is the Plücker coordinate of the rotational axis $\boldsymbol{n}$.

We next define the HDQ as

$$\widehat{\boldsymbol{q}} = \left[C\left(\frac{\widehat{\theta}}{2} \right), S\left(\frac{\widehat{\theta}}{2} \right) \widehat{\boldsymbol{n}} \right] \tag{19}$$

where $\widehat{\boldsymbol{n}}$ is the hyper-dual vector first introduced in [33], and $\widehat{\theta}$ is the hyper-dual angle [28,29]:

$$\widehat{\theta} = \hat{\theta} + \varepsilon^* \dot{\hat{\theta}} = \theta + \varepsilon d + \varepsilon^* \left(\dot{\theta} + \varepsilon \dot{d} \right) \tag{20}$$

Applying the Taylor series expansion of a hyper-dual function [28,33], one obtains the following Sine and Cosine trigonometric functions of a hyper-dual angle:

$$\begin{aligned}\sin\left(\widehat{\theta}\right) &= \sin(\theta) + \varepsilon d\cos(\theta) + \varepsilon^*\left(\dot{\theta}\cos(\theta) + \varepsilon\left(\dot{d}\cos(\theta) - d\dot{\theta}\sin(\theta)\right)\right)\\ \cos\left(\widehat{\theta}\right) &= \cos(\theta) - \varepsilon d\sin(\theta) - \varepsilon^*\left(\dot{\theta}\sin(\theta) + \varepsilon\left(\dot{d}\sin(\theta) + d\dot{\theta}\cos(\theta)\right)\right)\end{aligned} \tag{21}$$

Consider a particular HDQ that represents a helical motion of a rigid body, for which $\widehat{\boldsymbol{n}} = n$ and $\dot{\boldsymbol{n}} = 0$, Eq. (19) takes the following form:

$$\begin{aligned}\widehat{\boldsymbol{q}} &= \left[C\left(\frac{\widehat{\theta}}{2}\right), S\left(\frac{\widehat{\theta}}{2}\right)\widehat{\boldsymbol{n}}\right] = \left[\cos\left(\frac{\theta}{2}\right) - \varepsilon\frac{d}{2}\sin\left(\frac{\theta}{2}\right) - \varepsilon^*\left(\frac{\dot{\theta}}{2}\sin\left(\frac{\theta}{2}\right) + \varepsilon\left(\frac{\dot{d}}{2}\sin\left(\frac{\theta}{2}\right) + \frac{d\dot{\theta}}{4}\cos\left(\frac{\theta}{2}\right)\right)\right),\right.\\ &\qquad \left.\left(\sin\left(\frac{\theta}{2}\right) + \varepsilon\frac{d}{2}\cos\left(\frac{\theta}{2}\right) + \varepsilon^*\left(\frac{\dot{\theta}}{2}\cos\left(\frac{\theta}{2}\right) + \varepsilon\left(\frac{\dot{d}}{2}\cos\left(\frac{\theta}{2}\right) - \frac{d\dot{\theta}}{4}\sin\left(\frac{\theta}{2}\right)\right)\right)\right)\widehat{\boldsymbol{n}}\right]\\ &= \left[\cos\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right)\boldsymbol{n}\right] + \varepsilon\left[-\frac{d}{2}\sin\left(\frac{\theta}{2}\right), \frac{d}{2}\cos\left(\frac{\theta}{2}\right)\boldsymbol{n}\right]\\ &\qquad + \varepsilon^*\left[-\left(\frac{\dot{\theta}}{2}\sin\left(\frac{\theta}{2}\right) + \varepsilon\left(\frac{\dot{d}}{2}\sin\left(\frac{\theta}{2}\right) + \frac{d\dot{\theta}}{4}\cos\left(\frac{\theta}{2}\right)\right)\right), \left(\frac{\dot{\theta}}{2}\cos\left(\frac{\theta}{2}\right) + \varepsilon\left(\frac{\dot{d}}{2}\cos\left(\frac{\theta}{2}\right) - \frac{d\dot{\theta}}{4}\sin\left(\frac{\theta}{2}\right)\right)\right)\boldsymbol{n}\right]\end{aligned} \tag{22}$$

where the hyper-dual part of Eq. (22) is a derivative with respect to time of the dual part of the hyper-dual quaternion.

3.1. Hyper-Dual Quaternion representation of rigid body kinematics

The hyper-dual quaternion encompasses the derivatives of a quaternion and hence can be utilized to obtain a rigid body's pose and its derivatives. Time derivative of a body pose is obtained from a dual quaternion using the dual twist as follows [34,35]:

$$\begin{aligned}\dot{\hat{\boldsymbol{q}}} &= \frac{1}{2}\hat{\boldsymbol{q}}\otimes\hat{\boldsymbol{\xi}}\\ \hat{\boldsymbol{\xi}} &= 2\hat{\boldsymbol{q}}^*\otimes\dot{\hat{\boldsymbol{q}}}\end{aligned} \tag{23}$$

where $\hat{\boldsymbol{\xi}}$ is the dual twist, a combination of rotational and translational velocities,

$$\hat{\boldsymbol{\xi}} = \boldsymbol{\omega} + \varepsilon\left(\boldsymbol{\omega}\times\boldsymbol{T} + \dot{\boldsymbol{T}}\right) \tag{24}$$

$\boldsymbol{\omega}$ is the rotational velocity about the instantaneous rotation axis and $\boldsymbol{T}$ is the point mass radius vector from this axis. We next define the HDQ to be expressed as:

$$\widehat{\boldsymbol{q}} = \hat{\boldsymbol{q}} + \varepsilon^*\dot{\hat{\boldsymbol{q}}} \tag{25}$$

Substituting Eqs. (8), (23) and (24) into Eq. (25), yields:

$$\begin{aligned}\widehat{\boldsymbol{q}} &= \hat{\boldsymbol{q}} + \varepsilon^*\frac{1}{2}\hat{\boldsymbol{q}}\otimes\hat{\boldsymbol{\xi}} = \boldsymbol{q} + \varepsilon\frac{1}{2}\boldsymbol{q}\otimes\boldsymbol{T} + \varepsilon^*\frac{1}{2}\left(\left(\boldsymbol{q} + \varepsilon\frac{1}{2}\boldsymbol{q}\otimes\boldsymbol{T}\right)\otimes\left(\boldsymbol{\omega} + \varepsilon\left(\boldsymbol{\omega}\times\boldsymbol{T} + \dot{\boldsymbol{T}}\right)\right)\right)\\ &= \boldsymbol{q} + \varepsilon\frac{1}{2}\boldsymbol{q}\otimes\boldsymbol{T} + \varepsilon^*\frac{1}{2}\left(\boldsymbol{q}\otimes\boldsymbol{\omega} + \varepsilon\left(\boldsymbol{q}\otimes\boldsymbol{\omega}\times\boldsymbol{T} + \boldsymbol{q}\otimes\dot{\boldsymbol{T}} + \frac{1}{2}\boldsymbol{q}\otimes\boldsymbol{T}\otimes\boldsymbol{\omega}\right)\right)\end{aligned} \tag{26}$$

Taking the product of multiple HDQ as for example in a serial robot's kinematic chain:

$$\begin{aligned}&\widehat{\boldsymbol{q}}_1\otimes\widehat{\boldsymbol{q}}_2 = \hat{\boldsymbol{q}}_1\otimes\hat{\boldsymbol{q}}_2 + \varepsilon^*\left(\hat{\boldsymbol{q}}_1\otimes\dot{\hat{\boldsymbol{q}}}_2 + \dot{\hat{\boldsymbol{q}}}_1\otimes\hat{\boldsymbol{q}}_2\right)\\ &\widehat{\boldsymbol{q}}_1\otimes\widehat{\boldsymbol{q}}_2\otimes\widehat{\boldsymbol{q}}_3 = \hat{\boldsymbol{q}}_1\otimes\hat{\boldsymbol{q}}_2\otimes\hat{\boldsymbol{q}}_3 + \varepsilon^*\left(\hat{\boldsymbol{q}}_1\otimes\hat{\boldsymbol{q}}_2\otimes\dot{\hat{\boldsymbol{q}}}_3 + \hat{\boldsymbol{q}}_1\otimes\dot{\hat{\boldsymbol{q}}}_2\otimes\hat{\boldsymbol{q}}_3 + \dot{\hat{\boldsymbol{q}}}_1\otimes\hat{\boldsymbol{q}}_2\otimes\hat{\boldsymbol{q}}_3\right)\\ &..\\ &....\\ &\widehat{\boldsymbol{q}}_1\otimes\widehat{\boldsymbol{q}}_2....\otimes\widehat{\boldsymbol{q}}_n = \hat{\boldsymbol{q}}_1\otimes\hat{\boldsymbol{q}}_2....\otimes\hat{\boldsymbol{q}}_n + \varepsilon^*\left(\dot{\hat{\boldsymbol{q}}}_1\otimes\hat{\boldsymbol{q}}_2\otimes\hat{\boldsymbol{q}}_3\otimes\hat{\boldsymbol{q}}_4...\otimes\hat{\boldsymbol{q}}_n + ... + \hat{\boldsymbol{q}}_1\otimes\hat{\boldsymbol{q}}_2....\otimes\dot{\hat{\boldsymbol{q}}}_{n-1}\otimes\hat{\boldsymbol{q}}_n + \hat{\boldsymbol{q}}_1\otimes\hat{\boldsymbol{q}}_2....\otimes\dot{\hat{\boldsymbol{q}}}_n\right)\end{aligned}$$

the hyper-dual part of $\widehat{\boldsymbol{q}}_1\otimes\widehat{\boldsymbol{q}}_2....\otimes\widehat{\boldsymbol{q}}_n$ is equal to:

$$\begin{aligned}&\textstyle\sum_{i=1}^{n}\left(\prod_{j=0}^{i-1}\left(\hat{\boldsymbol{q}}_j\right)\cdot\dot{\hat{\boldsymbol{q}}}_i\cdot\prod_{k=i+1}^{n}\left(\hat{\boldsymbol{q}}_k\right)\right)\\ &\qquad\hat{\boldsymbol{q}}_0 = 1\end{aligned} \tag{27}$$

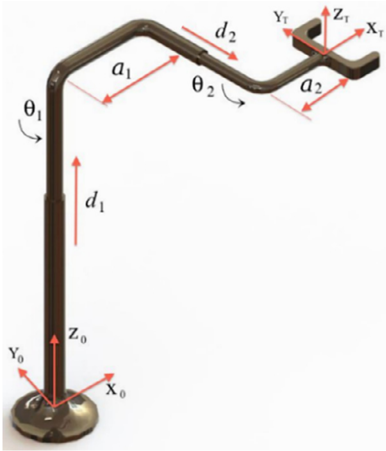

Fig. 2. Two cylindrical, four degrees-of-freedom robot.

which leads to the following recursive formula for n consecutive HDQ multiplications,

$$\overset{\frown}{\boldsymbol{q}}_f = \overset{\frown}{\boldsymbol{q}}_1 \otimes \overset{\frown}{\boldsymbol{q}}_2 .... \otimes \overset{\frown}{\boldsymbol{q}}_n = \hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{q}}_2 .... \otimes \hat{\boldsymbol{q}}_n + \varepsilon^* \left(\sum_{i=1}^{n} \left(\prod_{j=0}^{i-1} \left(\hat{\boldsymbol{q}}_j \right) \cdot \dot{\hat{\boldsymbol{q}}}_i \cdot \prod_{k=i+1}^{n} \left(\hat{\boldsymbol{q}}_k \right) \right) \right) \tag{28}$$

Where $\overset{\frown}{\boldsymbol{q}}_f$ is the resulting HDQ, and the hyper-dual part of the above equation is equal to the derivative of the dual part according to the chain rule, assuming all functions are differentiable.

Using Eqs. (23) and (25), the HDQ is expressed as:

$$\overset{\frown}{\boldsymbol{q}} = \hat{\boldsymbol{q}} + \varepsilon^* \frac{1}{2} \hat{\boldsymbol{q}} \otimes \hat{\boldsymbol{\xi}} \tag{29}$$

which leads to the flowing result for n consecutive HDQ multiplications,

$$\overset{\frown}{\boldsymbol{q}}_f = \hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{q}}_2 .... \otimes \hat{\boldsymbol{q}}_n + \varepsilon^* \frac{1}{2} \left(\hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{\xi}}_1 \otimes \hat{\boldsymbol{q}}_2 .... \otimes \hat{\boldsymbol{q}}_n + ... + \hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{q}}_2 .... \otimes \hat{\boldsymbol{q}}_{n-1} \otimes \hat{\boldsymbol{\xi}}_{n-1} \otimes \hat{\boldsymbol{q}}_n + \hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{q}}_2 .... \otimes \hat{\boldsymbol{q}}_n \otimes \hat{\boldsymbol{\xi}}_n \right) \tag{30}$$

The recursive formula for n consecutive HDQ transformations is, then, obtained by:

$$\overset{\frown}{\boldsymbol{q}}_f = \hat{\boldsymbol{q}}_1 \otimes \hat{\boldsymbol{q}}_2 .... \otimes \hat{\boldsymbol{q}}_n + \varepsilon^* \frac{1}{2} \left(\sum_{i=1}^{n} \left(\prod_{j=1}^{i} \left(\hat{\boldsymbol{q}}_j \right) \cdot \hat{\boldsymbol{\xi}}_i \cdot \prod_{k=i+1}^{n} \left(\hat{\boldsymbol{q}}_k \right) \right) \right) \tag{31}$$

Defining the product of n HDQ multiplications as

$$\overset{\frown}{\boldsymbol{q}}_f = \hat{\boldsymbol{q}}_{f\_d} + \varepsilon^* \hat{\boldsymbol{q}}_{f\_hd} \tag{32}$$

where $\hat{\boldsymbol{q}}_{f\_hd} = \dot{\hat{\boldsymbol{q}}}_{f\_d}$, then the resulting dual twist is given by:

$$\hat{\boldsymbol{\xi}}_f = 2 \hat{\boldsymbol{q}}^*_{f\_d} \otimes \hat{\boldsymbol{q}}_{f\_hd} \tag{33}$$

4. Illustrative example of a serial manipulator

Robot manipulators are an example of a multi-body system connected by joints. DQ are widely used in robotics (see for example [36–42]).

Consider the serial manipulator given in Fig. 2, also given in [28,29].

In [28] the derivation of the forward kinematic of a serial manipulator, was expressed as multiplications of hyper-dual 3 × 3 transformation matrices.

Similarly, in the hyper-dual quaternion form, by applying the hyper-dual angle and the extended principle of transference given in [33], the serial manipulator's kinematics is expressed as multiplications of HDQ as follows:

$$\overset{\frown}{\boldsymbol{q}}_{ef} = \overset{\frown}{\boldsymbol{q}}_1 \otimes \overset{\frown}{\boldsymbol{q}}_2 \otimes \overset{\frown}{\boldsymbol{q}}_3 \otimes \overset{\frown}{\boldsymbol{q}}_4 \tag{34}$$

where $\overset{\frown}{\boldsymbol{q}}_1$ to $\overset{\frown}{\boldsymbol{q}}_4$ are HDQ dual quaternions corresponding to each transformation, for which $\overset{\frown}{\boldsymbol{q}}_1$ represents a translation and rotation by θ_1 and d_1, $\overset{\frown}{\boldsymbol{q}}_2$ represents a constant translation by a_1, $\overset{\frown}{\boldsymbol{q}}_3$ represents a translation and rotation by θ_2 and d_2, and $\overset{\frown}{\boldsymbol{q}}_4$ represents a constant translation by a_2.

Keeping in mind that for the special case as a revolute or prismatic joint where the screw axis is coincident with the joint axis, one may write $\widehat{\boldsymbol{n}}_i = \boldsymbol{n}_i$, and Eq. (19) takes the form:

$$\widehat{\boldsymbol{q}}_i = \left[C\left(\frac{\widehat{\theta}_i}{2}\right), S\left(\frac{\widehat{\theta}_i}{2}\right)\boldsymbol{n}_i\right]$$

Hence from Eq. (22),

$$\begin{aligned}
\widehat{\boldsymbol{q}}_1 &= \left[C\left(\frac{\theta_1}{2}\right), S\left(\frac{\theta_1}{2}\right)[0,0,1]\right] + \varepsilon\left[-\frac{d_1}{2}S\left(\frac{\theta_1}{2}\right), \frac{d_1}{2}C\left(\frac{\theta_1}{2}\right)[0,0,1]\right] \\
&+ \varepsilon^*\left[-\left(\frac{\dot\theta_1}{2}S\left(\frac{\theta_1}{2}\right) + \varepsilon\left(\frac{\dot d_1}{2}S\left(\frac{\theta_1}{2}\right) + \frac{d_1\dot\theta_1}{4}C\left(\frac{\theta_1}{2}\right)\right)\right), \left(\frac{\dot\theta_1}{2}C\left(\frac{\theta_1}{2}\right) + \varepsilon\left(\frac{\dot d_1}{2}C\left(\frac{\theta_1}{2}\right) - \frac{d_1\dot\theta_1}{4}S\left(\frac{\theta_1}{2}\right)\right)\right)[0,0,1]\right] \\
\widehat{\boldsymbol{q}}_2 &= [1,0] + \varepsilon\left[0, \frac{a_1}{2}[1,0,0]\right] \\
\widehat{\boldsymbol{q}}_3 &= \left[C\left(\frac{\theta_2}{2}\right), S\left(\frac{\theta_2}{2}\right)[0,1,0]\right] + \varepsilon\left[-\frac{d_2}{2}S\left(\frac{\theta_2}{2}\right), \frac{d_2}{2}C\left(\frac{\theta_2}{2}\right)[0,1,0]\right] \\
&+ \varepsilon^*\left[-\left(\frac{\dot\theta_2}{2}S\left(\frac{\theta_2}{2}\right) + \varepsilon\left(\frac{\dot d_2}{2}S\left(\frac{\theta_2}{2}\right) + \frac{d_2\dot\theta_2}{4}C\left(\frac{\theta_1}{2}\right)\right)\right), \left(\frac{\dot\theta_2}{2}C\left(\frac{\theta_2}{2}\right) + \varepsilon\left(\frac{\dot d_2}{2}C\left(\frac{\theta_2}{2}\right) - \frac{d_2\dot\theta_2}{4}S\left(\frac{\theta_2}{2}\right)\right)\right)[0,1,0]\right] \\
\widehat{\boldsymbol{q}}_4 &= [1,0] + \varepsilon\left[0, \frac{a_2}{2}[1,0,0]\right]
\end{aligned}$$

for which $\widehat{\boldsymbol{q}}_{ef} = \hat{\boldsymbol{q}}_{ef\_d} + \varepsilon^*\hat{\boldsymbol{q}}_{ef\_hd}$, (see Appendix B for a complete definition of $\hat{\boldsymbol{q}}_{ef\_d}$ and $\hat{\boldsymbol{q}}_{ef\_hd}$).

Next by applying Eq. (33) one may now calculate the twist of the end-effector with no need for further differentiation yielding the rotational and translational velocities of the end-effector.

$$\begin{aligned}
\hat{\boldsymbol{\xi}}_{ef} &= 2\hat{\boldsymbol{q}}^*_{ef_d} \otimes \hat{\boldsymbol{q}}_{ef_hd} = \left[0, \boldsymbol{\omega}_{ef} + \varepsilon\boldsymbol{v}_{ef}\right] \\
&= \left[0, -\dot\theta_1 S(\theta_2), \dot\theta_2, \dot\theta_1 C(\theta_2)\right] + \varepsilon\left[0, -\dot d_1 S(\theta_2) - \dot\theta_1 d_2 C(\theta_2), \dot d_2 + \dot\theta_1 a_1 + \dot\theta_1 a_2 C(\theta_2), \dot d_1 C(\theta_2) - \dot\theta_2 a_2 - \dot\theta_1 d_2 S(\theta_2)\right]
\end{aligned}$$

It should be mentioned that the kinematic equations, Eq. (34), represented by HDQ, is compact and straightforward. The expansion however of this equation results in a complicated equation. From computational efficiency however this expression is more efficient and less computational intensive then homogeneous transformation matrices [3,43,44], and also spares the need for a separated velocity transformation calculate for example by the Jacobian matrices.

5. Conclusions

This paper introduces for the first time, the hyper-dual quaternions (HDQ). Hyper-dual quaternion is composed of hyper-dual numbers (HDN) as components of the quaternion, yielding a hyper-complex hyper-dual number. The arithmetic operations of HDQ is then developed and by using the hyper-dual angle as components of the HDQ it was applied to rigid body kinematics. It was shown that rigid body kinematics, that contains both translational and rotational motions and their derivatives, are obtained in a compact HDQ expression with no need for explicit differentiation.

This approach provides a systematic and compact means for the derivation of a single and multi-body systems' kinematics, kinematics terms i.e. body pose and its derivatives are embedded in the products of the HDQ, and there is no need to differentiate the expressions of the bodies' pose to obtain its derivatives.

A four degrees of freedom serial manipulator is used as an illustrative example of a multi-body system, for which the equations of motion in hyper-dual quaternion form are developed.

For acceleration, jerk, snap or higher order derivative calculations, one may expand the HDQ to dual-quaternions of higher orders, taking advantage of the extension of Kotelnikov's "Principle of transference" to dual-numbers of order n [33].

Appendix A

For a quaternion, $\boldsymbol{q} = [a, \boldsymbol{v}]$, where a is named the scalar part, $\boldsymbol{v}$ is named the vector part, the elementary arithmetic operations are:

Scalar Multiplication

$$s\boldsymbol{q} = [sa, s\boldsymbol{v}] \tag{A.1}$$

Addition

$$\boldsymbol{q}_1 + \boldsymbol{q}_2 = [a_1, \boldsymbol{v}_1] + [a_2, \boldsymbol{v}_2] = [a_1 + a_2, \boldsymbol{v}_1 + \boldsymbol{v}_2] \tag{A.2}$$

Quaternion Multiplication

$$\boldsymbol{q}_1 \otimes \boldsymbol{q}_2 = \left[a_1 a_2 - \boldsymbol{v}_1^T\boldsymbol{v}_2,\ a_1\boldsymbol{v}_2 + a_2\boldsymbol{v}_1 + \boldsymbol{v}_1 \times \boldsymbol{v}_2\right] \tag{A.3}$$

Conjugate

$$\boldsymbol{q}^* = [a, -\boldsymbol{v}] \tag{A.4}$$

Magnitude

$$\begin{aligned} \boldsymbol{q} &= a + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k} \\ \|\boldsymbol{q}\| &= \sqrt{\boldsymbol{q}\boldsymbol{q}^*} = \sqrt{a^2 + q_1^2 + q_2^2 + q_3^2} \end{aligned} \tag{A.5}$$

Unit Condition

$$\|\boldsymbol{q}\| = 1 \tag{A.6}$$

Appendix B

$$\begin{aligned}
\hat{\boldsymbol{q}}_{ef_d} = \bigg[& C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + \varepsilon\bigg(-\frac{d_1}{2}S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{d_2}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - \frac{a_1}{2}S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \frac{a_2}{2}S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg), \\
& - S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \varepsilon\bigg(\frac{a_2}{2}C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{d_2}{2}S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + \frac{a_1}{2}C\left(\frac{\theta_2}{2}\right)C\left(\frac{\theta_1}{2}\right) - \frac{d_1}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg), \\
& C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \varepsilon\bigg(\frac{d_2}{2}C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + \frac{a_2}{2}S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + \frac{a_1}{2}C\left(\frac{\theta_2}{2}\right)S\left(\frac{\theta_1}{2}\right) - \frac{d_1}{2}S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg), \\
& C\left(\frac{\theta_2}{2}\right)S\left(\frac{\theta_1}{2}\right) + \varepsilon\bigg(\frac{d_1}{2}C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + \frac{a_1}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - \frac{a_2}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - \frac{d_2}{2}S\left(\frac{\theta_2}{2}\right)S\left(\frac{\theta_1}{2}\right)\bigg)\bigg] \\
\hat{\boldsymbol{q}}_{ef_hd} = \bigg[& -\frac{\dot{\theta}_1}{2}S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{\dot{\theta}_2}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \varepsilon\bigg(-\frac{\dot{d}_1}{2}S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{\dot{d}_2}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \frac{\dot{\theta}_1}{4}\bigg(-d_1C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - a_1C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) \\
& + d_2S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + a_2C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg) + \frac{\dot{\theta}_2}{4}\bigg(d_1S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - a_1S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + d_2C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + a_2S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right)\bigg)\bigg), \\
& - S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{\dot{\theta}_1}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \varepsilon\bigg(-\frac{\dot{d}_1}{2}C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - \frac{\dot{d}_2}{2}C\left(\frac{\theta_2}{2}\right)S\left(\frac{\theta_1}{2}\right) + \frac{\dot{\theta}_1}{4}\bigg(d_1S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - a_1S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) \\
& - d_2C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - a_2S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right)\bigg) + \frac{\dot{\theta}_2}{4}\bigg(-d_1C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - a_1C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + d_2S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - a_2C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg)\bigg), \\
& - \frac{\dot{\theta}_1}{2}S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \varepsilon\bigg(-\frac{\dot{d}_1}{2}S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \frac{\dot{d}_2}{2}C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + \frac{\dot{\theta}_1}{4}\bigg(-d_1C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + a_1C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - d_2S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) \\
& + +a_2C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right)\bigg) + \frac{\dot{\theta}_2}{4}\bigg(-d_1S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - d_2C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - a_1S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - a_2S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg)\bigg), S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) \\
& + \frac{\dot{\theta}_1}{2}C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{\dot{\theta}_2}{2}S\left(\frac{\theta_2}{2}\right)S\left(\frac{\theta_1}{2}\right) + \varepsilon\bigg(\frac{\dot{d}_1}{2}C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - \frac{\dot{d}_2}{2}S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + \frac{\dot{\theta}_1}{4}\bigg(\bigg(-d_1S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - a_1S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) \\
& - d_2C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) + a_2S\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right)\bigg) + \frac{\dot{\theta}_2}{4}\bigg(-d_1C\left(\frac{\theta_1}{2}\right)S\left(\frac{\theta_2}{2}\right) - d_2S\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) + a_1C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right) - a_2C\left(\frac{\theta_1}{2}\right)C\left(\frac{\theta_2}{2}\right)\bigg)\bigg)\bigg]
\end{aligned}$$

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